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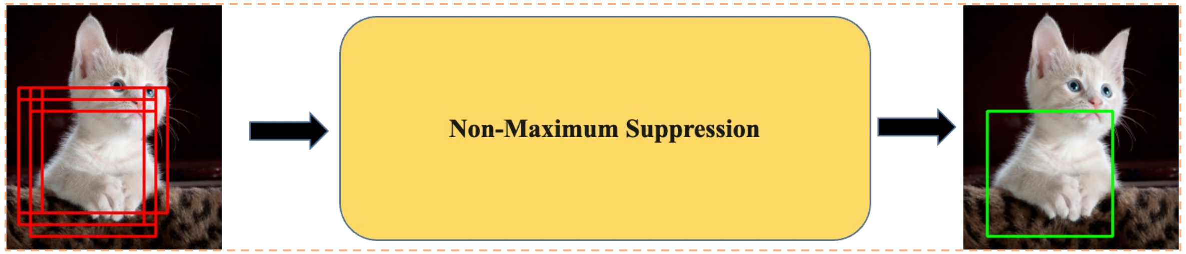
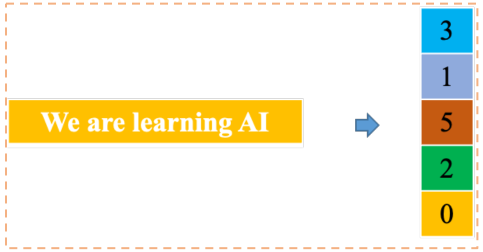
Data Structure

Tuple, Set and Dictionary



IOU, NMS using Tuple



Text embedding using Set

Histogram using Dictionary

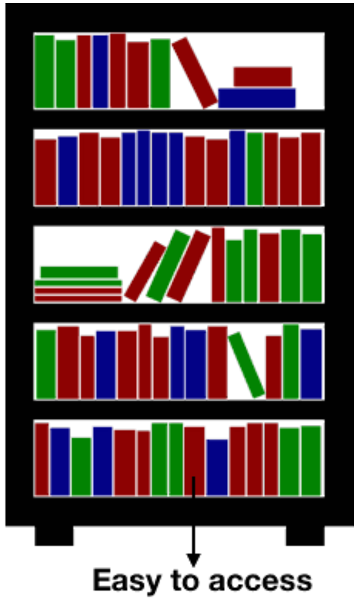
Vinh Dinh Nguyen
PhD in Computer Science

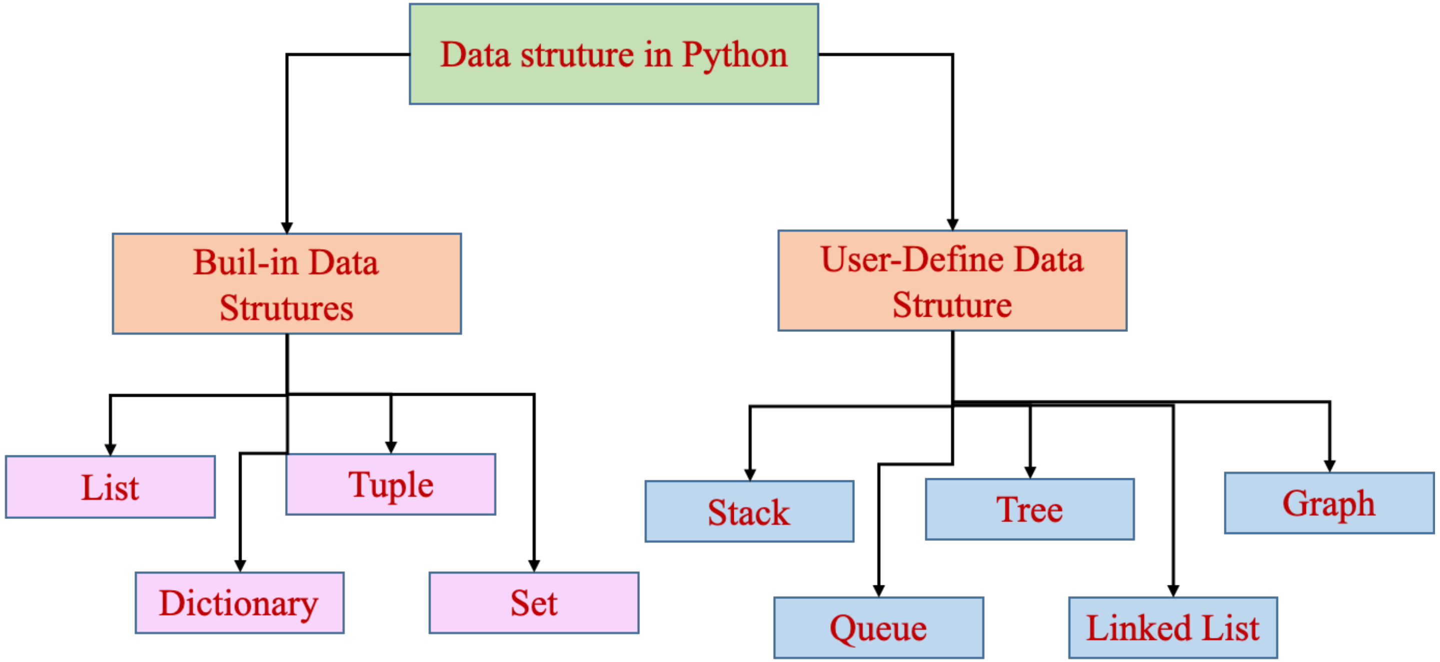
Outline

- What is a Tuple in Python
- What is a Set in Python
- What is a Dictionary in Python
- Quizz
- Text Classification using Set
- Image Histogram using Dictionary
- Further Reading

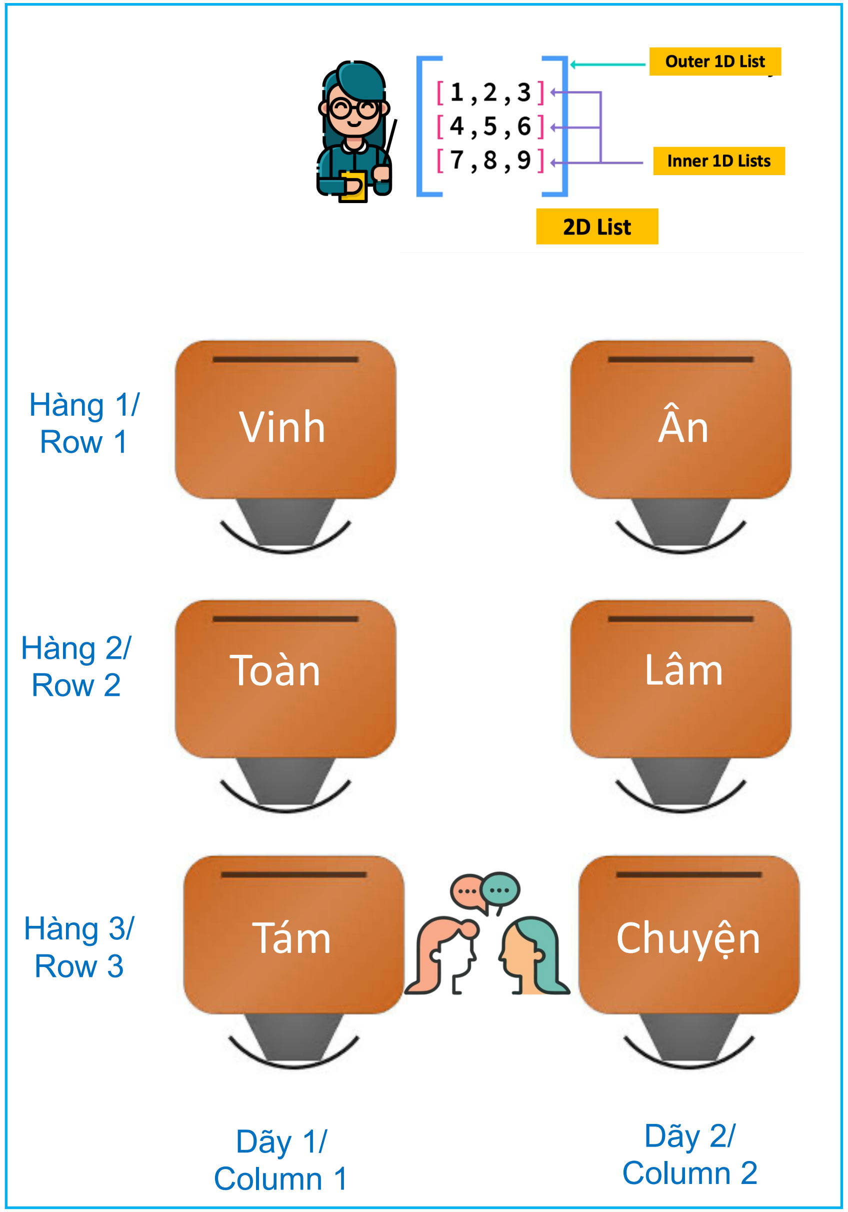
Data Structure?

A data structure is a storage that is used to store and organize data. It is a way of arranging data on a computer so that it can be accessed and updated efficiently.



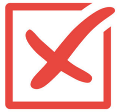


Review: From 1D to 2D List



Given that the teacher cannot remember students' names, please develop a program to help her find a student's name based on his/her seat location in the class.

"Cô mời 2 bạn: ngồi ở vị trí hàng thứ 3, dãy thứ 1, và ngồi ở vị trí hàng thứ 3, dãy thứ 2" lên bảng giải thích về 2D List



"Cô mời bạn Tám và bạn Chuyện" lên bảng giải thích về 2D List

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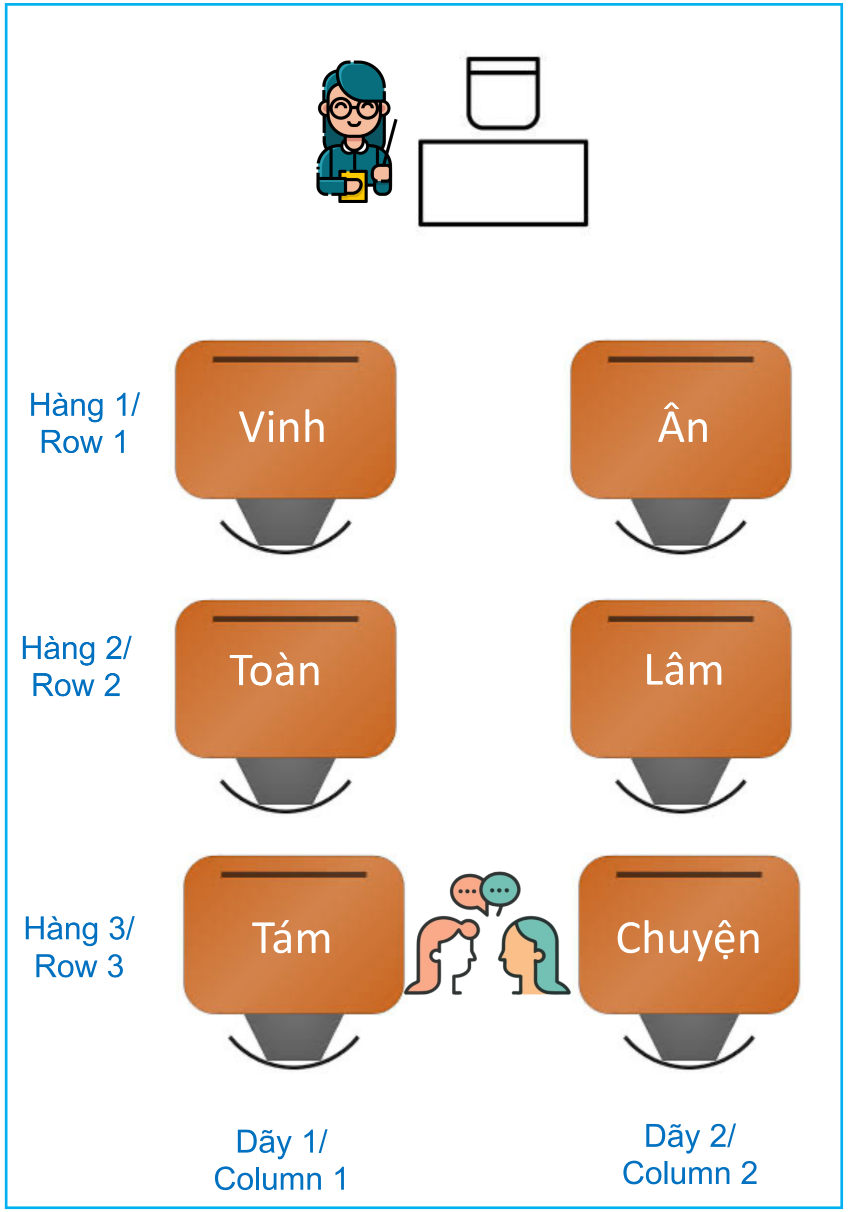
1D List Solution




```
student_list = ["Vinh", "An", "Toan", "Lam", "Tam", "Chuyen"]
row_class = 3
col_class = 2
def find_student(student_list, r, c):
    index = (r-1)*col_class + (c-1)
    return student_list[index]

#Find student a location (3, 1)
print(find_student(student_list, 3, 1))

#Find student a location (3, 2)
print(find_student(student_list, 3, 2))
```

```
Tam
Chuyen
```

Are there any easier solutions? This one is really difficult for newcomers

2D List Solution

Progrogramer View

	Column index 0	Column index 1
Row index 0	Vinh	An
Row index 1	Toan	Lam
Row index 2	Tam	Chuyen

Teacher View

	Column 1	Column 2
Row 1	Vinh	An
Row 2	Toan	Lam
Row 3	Tam	Chuyen

Use 1D List	Vinh	An
Use 1D List	Toan	Lam
Use 1D List	Tam	Chuyen

student_list = [element1, element2, element3)

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2D List Solution

Teacher View

	Column 1	Column 2
Row 1	Vinh	An
Row 2	Toan	Lam
Row 3	Tam	Chuyen

	Column index 0	Column index 1
Row index 0	Vinh	An
Row index 1	Toan	Lam
Row index 2	Tam	Chuyen

Progrogramer View

```
row1_student = ["Vinh", "An"] #1D List
row2_student = ["Toan", "Lam"] #1D List
row3_student = ["Tam", "Chuyen"] #1D List

student_list = [row1_student, row2_student, row3_student] # 2D List
```

How to access element:
row 0 and column 1
row 0 and column 2

row1_retrieve = student_list[0] →	Vinh, An
row1_retrieve[0] →	Vinh
row1_retrieve[1] →	An

student_list[0][0] →	Vinh
student_list[0]1] →	An

2D List Solution

	Column 1	Column 2
Row 1	Vinh	An
Row 2	Toan	Lam
Row 3	Tam	Chuyen

Teacher View

	Column index 0	Column index 1
Row index 0	Vinh	An
Row index 1	Toan	Lam
Row index 2	Tam	Chuyen

Progrogramer View

```
row1_student = ["Vinh", "An"] #1D List
row2_student = ["Toan", "Lam"] #1D List
row3_student = ["Tam", "Chuyen"] #1D List

student_list = [row1_student, row2_student, row3_student] # 2D List

num_row = len(student_list)
num_col = len(student_list[0])

def find_student(student_list, r, c):
    return student_list[r-1][c-1]

#Find student a location (3, 1)
print(find_student(student_list, 3, 1))

#Find student a location (3, 2)
print(find_student(student_list, 3, 2))
```

```
Tam
Chuyen
```

Thanks! This one is easy to understand

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Mutable Vs. Immutable

In Python, everything is treated as an object. Every object has these three attributes:

- ❑ Identity – This refers to the address that the object refers to in the computer's memory.
- ❑ Type – This refers to the kind of object that is created.
- ❑ Value – This refers to the value stored by the object.

Mutable	Immutable
We can change the value of a variable (which we have stored)	We can not change the value of a variable (which we have stored)
List, Dictionaries, Set, …	String, Tuples, …

While ID and Type cannot be changed once it's created, values can be changed for Mutable objects.

Mutable Vs. Immutable

❖List is Mutable

```
list=['a','b','c','d','e','f','g','h']
print("original list ")
print(list)

 # changing element at index 4 ,i.e., e to hello
list[4]='hello'
print("changed list")
print(list)
```

Output

```
original list
['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h']
changed list
['a', 'b', 'c', 'd', 'hello', 'f', 'g', 'h']
```

❖String is Immutable

```
words = "AI is trending"
words[len(words)-1] = "o"
print(words)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
<ipython-input-2-d66ddedf5312> in <cell line: 2>()
      1 words = "AI is trending"
----> 2 words[len(words)-1] = "o"
      3 print(words)

TypeError: 'str' object does not support item assignment
```

Mutable Vs. Immutable

❖List is Mutable

```python
# Create a string and a list
list1 = [1, 2, 3, 4, 5]
list2 = [1, 2, 3, 4, 5]

# Get the memory addresses
list1_address = id(list1)
list2_address = id(list2)

# Print the memory addresses
print(f"The memory address of the list 1 is: {list1_address}")
print(f"The memory address of the list 2 is: {list2_address}")
```

```
The memory address of the list 1 is: 137537838201792
The memory address of the list 2 is: 137538543554816
```

❖String is Immutable

```python
str1 = "AIO2024"
str2 = "AIO2024"

# Get the memory addresses
str1_address = id(str1)
str2_address = id(str2)

# Print the memory addresses
print(f"The memory address of the str1 is: {str1_address}")
print(f"The memory address of the str2 is: {str2_address}")
```

```
The memory address of the str1 is: 137537836287088
The memory address of the str2 is: 137537836287088
```

Object Interning In Python

Optimize memory usage

Tuple Motivation

❖ **Develop a function to return personal information (multiple values)**

```
# Function returning multiple values
def get_person():
    name = "Camedinh"
    age = 2004
    occupation = "Student"
    return [name, age, occupation]

# Test
person = get_person()
person[0] = ″ Your name has been hacked "
print(person[0], person[1], person[2])
```

List Example

```
Your name has been hacked, 2024, Student
```

List → **Tuple**

['Compile', 'With', 'Favtutor'] → ('Compile', 'With', 'Favtutor')

❖ **Can we prevent this happen?**

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Tuple Motivation

❖ **Develop a function to return personal information (multiple values)**

```
# Function returning multiple values
def get_person():
    name = "Camedinh"
    age = 2004
    occupation = "Student"
    return (name, age, occupation)

# Unpacking the tuple
person = get_person()
person[0] = "Your name is Hack"
print(person[0], person[1], person[2])
```

Tuple Example

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
<ipython-input-22-3693fe48cbb2> in <cell line: 10>()
      8 # Unpacking the tuple
      9 person = get_person()
---> 10 person[0] = "Your name is Hack"
     11 print(person[0], person[1], person[2])

TypeError: 'tuple' object does not support item assignment
```

❖ **How to use a Tuple**

Tuples Are Like Lists

Tuples are another kind of sequence that functions much like a list - they have elements which are indexed starting at 0

```
>>> x = ('Glenn', 'Sally', 'Joseph')
>>> print(x[2])
Joseph
>>> y = ( 1, 9, 2 )
>>> print(y)
(1, 9, 2)
>>> print(max(y))
9
```

```
>>> for iter in y:
...     print(iter)
...
1
9
2
>>>
```

What is a Tuple

but... Tuples are “immutable”

Unlike a list, once you create a tuple, you cannot alter its contents - similar to a string

```
>>> x = [9, 8, 7]
>>> x[2] = 6
>>> print(x)
>>>[9, 8, 6]
>>>
```

```
>>> y = 'ABC'
>>> y[2] = 'D'
Traceback:'str' object does
not support item
Assignment
>>>
```

```
>>> z = (5, 4, 3)
>>> z[2] = 0
Traceback:'tuple' object does
not support item
Assignment
>>>
```

What is a Tuple

Built-in Functions in List and Tuple

```
>>> l = list()
>>> dir(l)
['append', 'count', 'extend', 'index', 'insert', 'pop', 'remove', 'reverse', 'sort']

>>> t = tuple()
>>> dir(t)
['count', 'index']
```

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What is a Tuple

❖ Structure

tuple_name = (element-1, …, element-n)

```
t = (1, 2, 3)

print(t[0])
print(t[1])
print(t[2])
```

```
1
2
3
```

Tuple unpacking

```
x1,y1,z1     = ('a','b','c')
(x2,y2,z2) = ('a','b','c')
print(x1)
print(x2)
```

```
a
a
```

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What is a Tuple

❖ Structure

tuple_name = (element-1, …, element-n)

( ) can be removed

```
t = 1, 2
print(t)
```

```
(1, 2)
```

Tuple with one element

```
var1 = (1 + 2) * 5
print(type(var1), ' ', var1)

var2 = (1)
print(type(var2), ' ', var2)

var3 = (1,)
print(type(var3), ' ', var3)
```

```
<class 'int'>    15
<class 'int'>    1
<class 'tuple'>    (1,)
```

What is a Tuple

❖ + and * operators

```python
t1 = (1, 0)
print(t1)

t1 += (2,)
print(t1)
```

```
(1, 0)
(1, 0, 2)
```

```python
t = (1,) * 5
```

```
(1, 1, 1, 1, 1)
```

count() - đếm số lần xuất hiện của một giá trị

index() - tìm vị trí xuất hiện của một giá trị

```python
t = (1,2,3,1)
count = t.count(1)
index = t.index(2)

print(count)
print(index)
```

```
2
1
```

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What is a Tuple

len() - Tìm chiều dài của một tuple

```
t = (1, 2, 3, 4)
len(t)
```

```
4
```

Lấy giá trị min và max của một tuple

```
t = (1, 2, 3, 4, 5)

print(min(t))
print(max(t))
print(sum(t))
```

```
1
5
15
```

Dùng hàm zip() cho tuple

```
t1 = (1, 2, 3, 4, 5)
t2 = ('a', 'b', 'c', 'd', 'e')

print(t1)
print(t2)

t3 = zip(t1, t2)
for x,y in t3:
    print(x, y)
```

```
(1, 2, 3, 4, 5)
('a', 'b', 'c', 'd', 'e')
1 a
2 b
3 c
4 d
5 e
```

Sắp xếp các giá trị trong một tuple

```
t = (4, 7, 3, 9, 6)
t_s = sorted(t)
print(t_s)
```

```
[3, 4, 6, 7, 9]
```

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Tuple Examples

Swapping two variables

```
def swap(v1, v2):
    (v2, v1) = (v1, v2)
    return (v1, v2)
```

```
v1 = 2
v2 = 3
(v1, v2) = swap(v1, v2)

# print
print(v1)
print(v2)
```

```
3
2
```

Memory requirement

```
# memory comparison
import sys

aList  = [3, 4, 5, 6, 7]
aTuple = (3, 4, 5, 6, 7)

print(sys.getsizeof(aList))
print(sys.getsizeof(aTuple))
```

```
120
80
```

Tuple slicing

```
data = (1, 2, 3, 4, 5)
print(data[2:])
print(data[::-1])
```

```
(3, 4, 5)
(5, 4, 3, 2, 1)
```

list2tuple

```
# convert from list to tuple
aList  = [3, 4, 5, 6, 7]
aTuple = tuple(aList)

print(aTuple)
print(type(aTuple))
```

```
(3, 4, 5, 6, 7)
<class 'tuple'>
```

tuple2list

```
# convert from tuple to list
aTuple  = (3, 4, 5, 6, 7)
aList = list(aList)

print(aList)
print(type(aList))
```

```
[3, 4, 5, 6, 7]
<class 'list'>
```

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Tuple Examples

❖ Example: Solve quadratic equation

```python
import math

def quadratic_equation(a, b, c):
    '''
    This function aims at solving the quadratic equation
    a, b, c --- three parameters and a =! 0
    '''

    # compute delta
    delta = b*b - 4*a*c

    if delta < 0:
        return ()
    elif delta == 0:
        x = (-b+math.sqrt(delta))/2*a
        return (x,)
    else:
        x1 = (-b+math.sqrt(delta))/(2*a)
        x2 = (-b-math.sqrt(delta))/(2*a)
        return (x1,x2)
```

Case 1: delta<0

```python
result = quadratic_equation(a=5, b=0, c=1)
print(type(result))
print(len(result))
print(result)
```

```
<class 'tuple'>
0
()
```

Case 2: delta>0

```python
result = quadratic_equation(a=5, b=5, c=1)
print(type(result))
print(len(result))
print(result)
```

```
<class 'tuple'>
2
(-0.276393202250021, -0.7236067977499789)
```

Case 3: delta=0

```python
result = quadratic_equation(a=4, b=4, c=1)
print(type(result))
print(len(result))
print(result)
```

```
<class 'tuple'>
1
(-8.0,)
```

Data is protected

```python
result = quadratic_equation(a=4, b=4, c=1)
result[0] = 1
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most
<ipython-input-21-55f394e5ddc8> in <module>
      1 result = quadratic_equation(a=4, b=4, c=1)
----> 2 result[0] = 1

TypeError: 'tuple' object does not support item assignment
```

Q&A Time

Outline

- What is a Tuple in Python
- What is a Set in Python
- What is a Dictionary in Python
- Quizz
- Text Classification using Set
- Image Histogram using Dictionary
- Further Reading

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Set Motivation

List

1	**0**	**2**	**1**	**0**	**2**

Set

1	3	2	4	5	6

Lists and tuples are standard Python data types that store values in a sequence. Sets are another standard Python data type that also store values. The major difference is that sets, unlike lists or tuples, cannot have multiple occurrences of the same element and store unordered values.

Set

❖ Create a set

Using curly brackets

```
# create a set
animals = {"cat", "dog", "tiger"}

print(type(animals))
print(animals)
```

```
<class 'set'>
{'dog', 'cat', 'tiger'}
```

Items with different data types

```
# create a set
a_set = {"cat", 5, True, 40.0}

print(type(a_set))
print(a_set)
```

```
<class 'set'>
{40.0, 'cat', 5, True}
```

Set comprehension

```
# set comprehension

a_set = {i*i for i in range(10)}
print(a_set)
```

```
{0, 1, 64, 4, 36, 9, 16, 49, 81, 25}
```

Set

Access the items of a set

```python
# accessing items
animals = {"cat", "dog", "tiger"}
for animal in animals:
    print(animal)
```

```
dog
cat
tiger
```

Copy a set

```python
# copy
animals = {"cat", "dog", "tiger"}
print("Animals:", animals)

a_copy = animals.copy()
print("Copy:", a_copy)
```

```
Animals: {'dog', 'cat', 'tiger'}
Copy: {'dog', 'cat', 'tiger'}
```

Set

Add an item

```
# add an item
animals = {"cat", "dog", "tiger"}
animals.add("bear")
print(animals)
```

```
{'dog', 'bear', 'cat', 'tiger'}
```

Insert a set to another set

```
# insert a set to another set
animals = {"cat", "dog", "tiger"}
animals.update({"chicken","Duck"})
print(animals)
```

```
{'Duck', 'tiger', 'dog', 'cat', 'chicken'}
```

Join two sets

```
# join two sets
set1 = {"cat", "dog"}
set2 = {"duck", "tiger"}

set3 = set1.union(set2)
print(set3)
```

```
{'duck', 'dog', 'cat', 'tiger'}
```

Not allow duplicate values

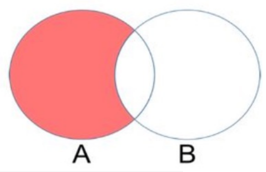
```
# No duplication
animals = {"cat", "dog", "tiger"}
print(animals)

animals.add("cat")
print(animals)
```

```
{'tiger', 'cat', 'dog'}
{'tiger', 'cat', 'dog'}
```

Set

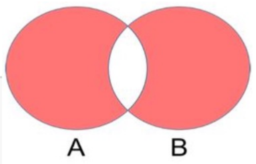
difference function

```
# difference

set1 = {"apple", "banana", "cherry"}
set2 = {"pineapple", "apple"}

set3 = set1.difference(set2)

print(set3)
```




```
{'cherry', 'banana'}
```

symmetric_difference




```
# symmetric_difference

set1 = {"apple", "banana", "cherry"}
set2 = {"pineapple", "apple"}

set3 = set1.symmetric_difference(set2)

print(set3)
```

```
{'pineapple', 'cherry', 'banana'}
```

difference_update function

```
# difference_update

set1 = {"apple", "banana", "cherry"}
set2 = {"pineapple", "apple"}

set1.difference_update(set2)

print(set1)
```

```
{'cherry', 'banana'}
```

symmetric_difference_update

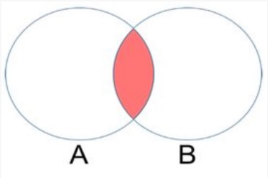
```
# symmetric_difference_update

set1 = {"apple", "banana", "cherry"}
set2 = {"pineapple", "apple"}

set1.symmetric_difference_update(set2)

print(set1)
```

```
{'pineapple', 'cherry', 'banana'}
```

Set

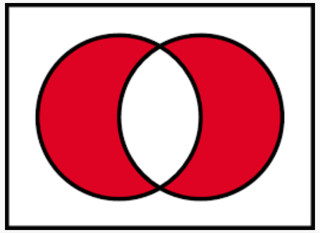
❖ Bitwise operator

```
# AND (&)

set1 = {1, 2, 3}
set2 = {3, 4, 5}

print(set1 & set2)
```



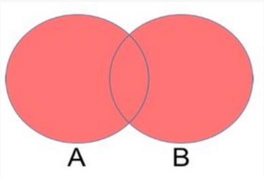

```
{3}
```

```
# XOR (^)

set1 = {1, 2, 3}
set2 = {3, 4, 5}

print(set1 ^ set2)
```



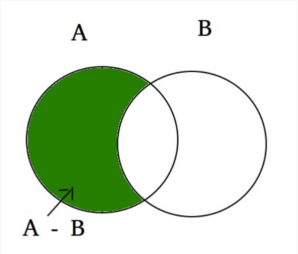
```
{1, 2, 4, 5}
```

```
# OR (|)

set1 = {1, 2, 3}
set2 = {3, 4, 5}

print(set1 | set2)
```




```
{1, 2, 3, 4, 5}
```

```
# subtraction (-)

set1 = {1, 2, 3}
set2 = {3, 4, 5}

print(set1 - set2)
```




```
{1, 2}
```

Set

❖ Remove an item

remove(item)

Remove an item from the set.

```
# remove an item
animals = {"cat", "dog", "tiger"}
animals.remove("dog")
print(animals)
```

```
{'cat', 'tiger'}
```

discard(item)

Remove an item from the set if it is present.

```
# remove an item
animals = {"cat", "dog", "tiger"}
animals.discard("tiger")
print(animals)
```

```
{'dog', 'cat'}
```

Set comprehension

```
# set comprehension

aSet = {i*i for i in range(10)}
print(aSet)
```

```
{0, 1, 64, 4, 36, 9, 16, 49, 81, 25}
```

https://docs.python.org/3/library/stdtypes.html?t#set

Set

❖ Remove an item

remove(item)

Remove an item from the set.

Raises KeyError if elem is not contained in the set.

```python
# remove an item
animals = {"cat", "dog", "tiger"}
animals.remove("duck")
print(animals)
```

```
---------------------------------------
KeyError
<ipython-input-29-604e724f3515> in <module
      1 # remove an item
      2 animals = {"cat", "dog", "tiger"}
----> 3 animals.remove("duck")
      4 print(animals)

KeyError: 'duck'
```

discard(item)

Remove an item from the set if it is present.

```python
# remove an item
animals = {"cat", "dog", "tiger"}
animals.discard("duck")
print(animals)
```

```
{'dog', 'cat', 'tiger'}
```

https://docs.python.org/3/library/stdtypes.html?t#set

Set

❖ Create a set

Unordered and unindexed

```
# not support indexing
animals = {"cat", "dog", "tiger"}
print(animals[1])
```

```
---------------------------------------------------------------
TypeError                                 Traceback
<ipython-input-24-a1b489f88deb> in <module>
      1 # not support indexing
      2 animals = {"cat", "dog", "tiger"}
----> 3 print(animals[1])

TypeError: 'set' object is not subscriptable
```

Cannot contain unhashable types

```
# create a set
a_list = [1, 2, 3]
a_set  = {"cat", a_list}
print(a_set)
```

```
---------------------------------------------------------------
TypeError                                 Traceback
<ipython-input-5-35efb5d3ad33> in <module>
      1 # shallow copy
      2 a_list = [1, 2, 3]
----> 3 a_set  = {"cat", a_list}
      4 print(a_set)

TypeError: unhashable type: 'list'
```

Set

Set ←→ List and Tuple

```python
# convert from set to list
aSet = {1, 2, 3, 4, 5}

aList = list(aSet)
print(aList)
print(type(aList))
```

```
[1, 2, 3, 4, 5]
<class 'list'>
```

```python
# convert from list to set
aList = [1, 2, 3, 2, 1]

aSet = set(aList)
print(aSet)
print(type(aSet))
```

???

```python
# convert from set to tuple
aSet = {1, 2, 3, 4, 5}

aTuple = tuple(aSet)
print(aTuple)
print(type(aTuple))
```

```
(1, 2, 3, 4, 5)
<class 'tuple'>
```

```python
# convert from tuple to set
aTuple = (1, 2, 3, 2, 1)

aSet = set(aTuple)
print(aSet)
print(type(aSet))
```

???

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Q&A Time

Outline

- What is a Tuple in Python
- What is a Set in Python
- What is a Dictionary in Python
- Quizz
- Text Classification using Set
- Image Histogram using Dictionary
- Further Reading

Dictionary Motivation

number_list = [11, 15, 35, 20, 18, 9]

	11	15	35	20	18	9
Index	0	1	2	3	4	5

index

value

number_list[0] => 11

It's very hard to remember that the 1st element in a list has index 0, is there an easier way to remember this?

number_dict ={"One":11, "Two":15, "Three":35, "Four":20, "Five":18, "Six:9" }

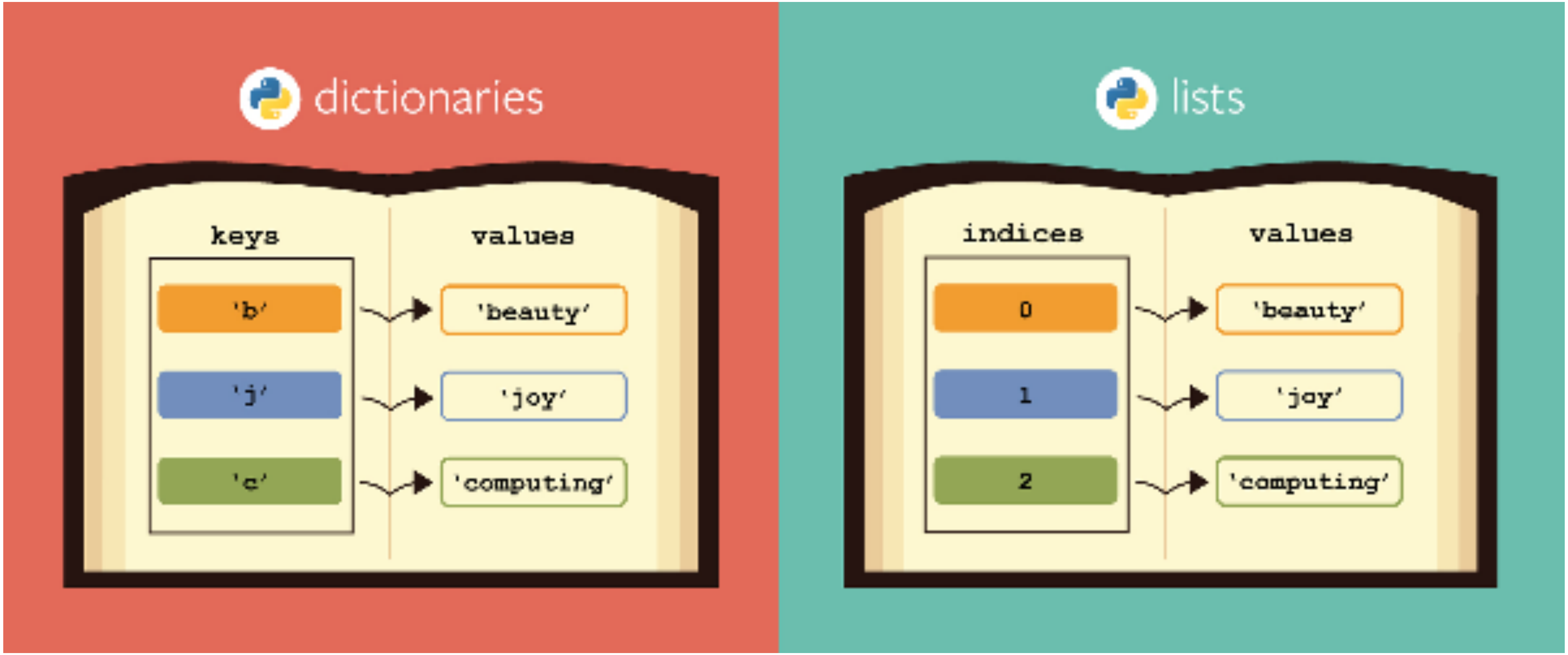
	11	15	35	20	18	9
key	"One	Two	Three	Four	Five	Six

key

value

number_dict["One"] => 11

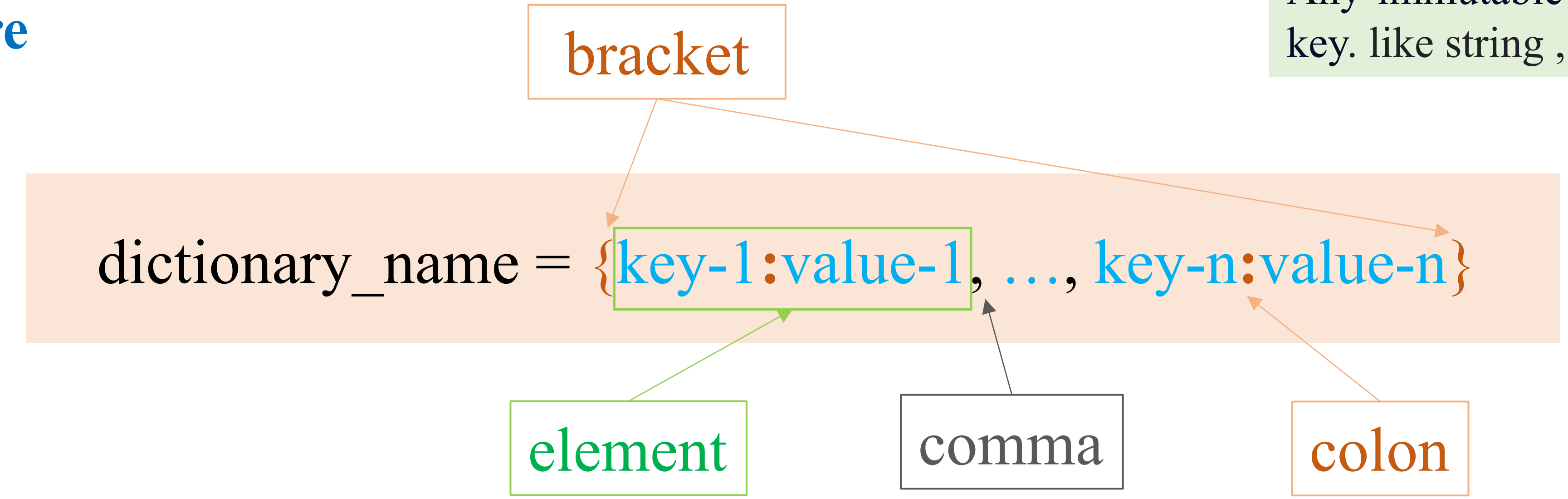
Dictionary Motivation



Dictionary

❖ Structure

Any immutable can be dictionary key. like string , number, tuple



Create a dictionary

```
parameters = {'learning_rate': 0.1,
              'optimizer': 'Adam',
              'metric': 'Accuracy'}

print(parameters)
print(type(parameters))
```

```
{'learning_rate': 0.1, 'optimizer': 'Adam', 'metric': 'Accuracy'}
<class 'dict'>
```

Comparing Lists and Dictionaries

Dictionaries are like lists except that they use keys instead of numbers to look up values

```
>>> lst = list()
>>> lst.append(21)
>>> lst.append(183)
>>> print(lst)
[21, 183]
>>> lst[0] = 23
>>> print(lst)
[23, 183]
```

```
>>> ddd = dict()
>>> ddd['age'] = 21
>>> ddd['course'] = 182
>>> print(ddd)
{'course': 182, 'age': 21}
>>> ddd['age'] = 23
>>> print(ddd)
{'course': 182, 'age': 23}
```

Dictionary

❖ Create a Dictionary

```python
# dic comprehension

a_dict = {str(i):i for i in range(5)}
print(a_dict)
```

```
{'0': 0, '1': 1, '2': 2, '3': 3, '4': 4}
```

```python
# from zip

tuple1 = (1, 2, 3)
tuple2 = (4, 5, 6)

a_dict = dict(zip(tuple1, tuple2))
print(type(a_dict))
print(a_dict)
```

```
<class 'dict'>
{1: 4, 2: 5, 3: 6}
```

```python
# from zip

set1 = {1, 2, 3}
set2 = {4, 5, 6}

a_dict = dict(zip(set1, set2))
print(type(a_dict))
print(a_dict)
```

```
<class 'dict'>
{1: 4, 2: 5, 3: 6}
```

```python
# from zip

list1 = [1, 2, 3]
list2 = [4, 5, 6]

a_dict = dict(zip(list1, list2))
print(type(a_dict))
print(a_dict)
```

```
<class 'dict'>
{1: 4, 2: 5, 3: 6}
```

Dictionary

❖ Update a value

```
parameters = {'learning_rate': 0.1,
              'metric': 'Accuracy'}

parameters['learning_rate'] = 0.2
print(parameters)
```

```
{'learning_rate': 0.2, 'metric': 'Accuracy'}
```

❖ Copy a dictionary

```
parameters = {'learning_rate': 0.1,
              'metric': 'Accuracy'}

a_copy = parameters.copy()
a_copy['learning_rate'] = 0.2

print(parameters)
print(a_copy)
```

```
{'learning_rate': 0.1, 'metric': 'Accuracy'}
{'learning_rate': 0.2, 'metric': 'Accuracy'}
```

Dictionary

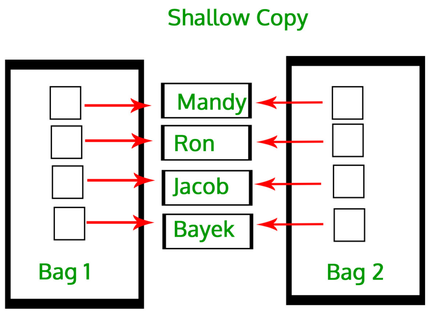
❖ Hàm copy() chỉ sao chép kiểu shallow

```
d1 = {'a': [1,2], 'b': 5}
d2 = d1.copy()

# thay đổi giá trị d2 sẽ ảnh hưởng đến d1
d2['a'][0] = 3
d2['a'][1] = 4

print('d1:', d1)
print('d2:', d2)
```

```
d1: {'a': [3, 4], 'b': 5}
d2: {'a': [3, 4], 'b': 5}
```



❖ Sử dụng hàm deepcopy() trong module copy

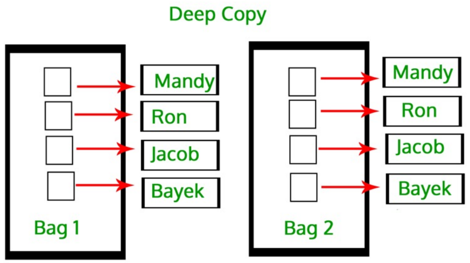
```
import copy

d1 = {'a': [1,2], 'b': 5}
d2 = copy.deepcopy(d1)

# thay đổi giá trị d2
d2['a'][0] = 3
d2['a'][1] = 4

print('d1:', d1)
print('d2:', d2)
```

```
d1: {'a': [1, 2], 'b': 5}
d2: {'a': [3, 4], 'b': 5}
```



Dictionary

❖ Get keys and values

Get keys

```python
keys = parameters.keys()
for key in keys:
    print(key)
```

```
learning_rate
optimizer
metric
```

Get values

```python
values = parameters.values()
for value in values:
    print(value)
```

```
0.1
Adam
Accuracy
```

Get keys

```python
parameters = {'learning_rate': 0.1,
              'optimizer': 'Adam',
              'metric': 'Accuracy'}

for key in parameters:
    print(key)
```

```
learning_rate
optimizer
metric
```

Get keys and values

```python
parameters = {'learning_rate': 0.1,
              'optimizer': 'Adam',
              'metric': 'Accuracy'}

items = parameters.items()
for key, value in items:
    print(key, value)
```

```
learning_rate 0.1
optimizer Adam
metric Accuracy
```

Dictionary

❖ Get a value by a key

Get value using get() function

```python
parameters = {'learning_rate': 0.1,
              'optimizer': 'Adam',
              'metric': 'Accuracy'}

value = parameters.get('learning_rate')
print(value)

print('\nAfter using get() function')
print(parameters)
```

```
0.1

After using get() function
{'learning_rate': 0.1, 'optimizer': 'Adam', 'metric': 'Accuracy'}
```

Get value and delete the corresponding item

```python
parameters = {'learning_rate': 0.1,
              'optimizer': 'Adam',
              'metric': 'Accuracy'}

value = parameters.pop('learning_rate')
print(value)

print('\nAfter using pop() function')
print(parameters)
```

```
0.1

After using pop() function
{'optimizer': 'Adam', 'metric': 'Accuracy'}
```

Dictionary

popitem() - lấy ra một phần tử ở cuối dictionary

```
parameters = {'learning_rate': 0.1,
              'optimizer': 'Adam',
              'metric': 'Accuracy'}

item = parameters.popitem()

print(item)
print(parameters)
```

```
('metric', 'Accuracy')
{'learning_rate': 0.1, 'optimizer': 'Adam'}
```

clear() - xóa tất cả các phần tử của một dictionary

```
parameters = {'learning_rate': 0.1,
              'metric': 'Accuracy'}

print('Before using clear() function')
print(parameters)

parameters.clear()

print('\nAfter using clear() function')
print(parameters)
```

```
Before using clear() function
{'learning_rate': 0.1, 'metric': 'Accuracy'}

After using clear() function
{}
```

Use del keyword to delete an item

```
parameters = {'learning_rate': 0.1,
              'metric': 'Accuracy'}
print(parameters)

del parameters['metric']
print(parameters)
```

```
{'learning_rate': 0.1, 'metric': 'Accuracy'}
{'learning_rate': 0.1}
```

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Dictionary

❖ Key that does not exist

Try to delete a non-existing item

```python
parameters = {'learning_rate': 0.1,
              'metric': 'Accuracy'}

del parameters['algorithm']
```

```
---------------------------------------------------------------------
KeyError                                  Traceback
<ipython-input-29-e759e1753a28> in <module>
      2               'metric': 'Accuracy'}
      3
----> 4 del parameters['algorithm']

KeyError: 'algorithm'
```

Try to get an item by a non-existing key

```python
parameters = {'learning_rate': 0.1,
              'optimizer': 'Adam',
              'metric': 'Accuracy'}

value = parameters.pop('algorithm')
```

```
---------------------------------------------------------------------
KeyError                                  Traceback
<ipython-input-30-8310550c04f4> in <module>
      3               'metric': 'Accuracy'}
      4
----> 5 value = parameters.pop('algorithm')

KeyError: 'algorithm'
```

Dictionary

setdefault() function

```
# setdefault()

fruits = {'banana': 2}
fruits.setdefault('apple', 0)

print(fruits)
```

```
{'banana': 2, 'apple': 0}
```

```
# setdefault()

fruits = {'banana': 2, 'apple': 4}
fruits.setdefault('apple', 0)

print(fruits)
```

```
{'banana': 2, 'apple': 4}
```

example

```
# setdefault()

fruits = {'banana': 2}
fruits.setdefault('apple', 0)

fruits['apple'] += 10
print(fruits)
```

```
{'banana': 2, 'apple': 10}
```

Result ???

```
# setdefault()

fruits = {'banana': 2}

fruits['apple'] += 10
print(fruits)
```

Dictionary

❖ Get a value via a key

Method 1

```
# access value via key

fruits = {'banana': 2, 'apple': 4}
print(fruits['apple'])
print(fruits['corn'])
```

```
4

---------------------------------------------------------------------------
KeyError                                  Traceback
<ipython-input-21-bda9d1f64a8f> in <module>
      3 fruits = {'banana': 2, 'apple': 4}
      4 print(fruits['apple'])
----> 5 print(fruits['corn'])

KeyError: 'corn'
```

Method 2

```
# access value via key

fruits = {'banana': 2, 'apple': 4}
print(fruits.get('apple'))
print(fruits.get('corn'))
```

```
4
None
```

Dictionary

Merge two dictionaries

```
# merge two dicts

fruits = {'banana': 2, 'apple': 4}
cereal = {'rice': 3, 'corn': 7}

result = {**fruits, **cereal}
print(result)
```

```
{'banana': 2, 'apple': 4, 'rice': 3, 'corn': 7}
```

Check if a key exists

```
# check if a key exists

fruits = {'banana': 2, 'apple': 4}

print('apple' in fruits)
print('corn' in fruits)
```

```
True
False
```

Remove empty items

```
# remove empty items

fruits = {'banana': 2, 'apple': None}

dict1 = {key:value for (key, value)
                   in fruits.items()
                   if value is not None}
print(dict1)
```

```
{'banana': 2}
```

Dictionary comprehension

```
# dic comprehension

aDict = {str(i):i for i in range(5)}
print(aDict)
```

```
{'0': 0, '1': 1, '2': 2, '3': 3, '4': 4}
```

Q&A Time

Outline

- What is a Tuple in Python
- What is a Set in Python
- What is a Dictionary in Python
- Quizz
- Text Classification using Set
- Image Histogram using Dictionary
- Further Reading

Outline

- What is a Tuple in Python
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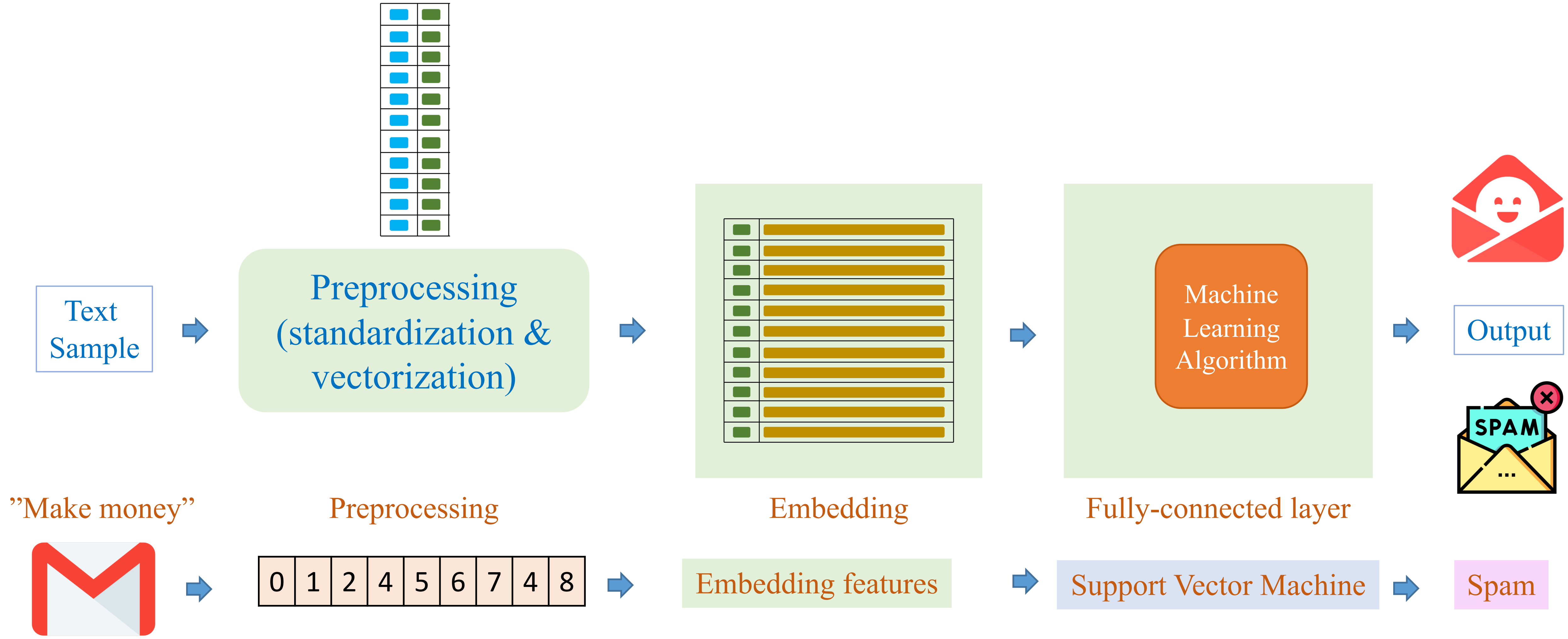
Set in Text Classification

❖ Text classification



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Set in Text Classification

- Example corpus
  sample1: ‘We are learning AI’
  sample2: ‘AI is a CS topic’

(1) Build vocabulary from corpus

index	0	1	2	3	4	5	6	7
word	pad	are/a	ai	we	topic	learning	is	cs

(2) Transform text into features

We are learning AI			
we	are	learning	ai
3	1	5	2

AI is a CS topic				
ai	is	a	cs	topic
2	6	1	7	4

Standardize, Vectorization

Set in Text Classification

❖Vocabulary Building

Python is a high-level, interpreted, general-purpose programming language. Its design philosophy emphasizes code readability with the use of significant indentation. Python is dynamically-typed and garbage-collected. It supports multiple programming paradigms, including structured, object-oriented and functional programming.

Python Program (using Set structure)

0 its	18 general
1 indentation	19 with
2 collected	20 philosophy
3 supports	21 object
4 is	22 level
5 programming	23 paradigms
6 garbage	24 design
7 purpose	25 a
8 it	26 the
9 including	27 oriented
10 and	28 emphasizes
11 typed	29 high
12 interpreted	30 significant
13 dynamically	31 structured
14 multiple	32 language
15 python	33 code
16 functional	34 use
17 readability	35 of

Vocabulary

Input

"Python is a high-level"

Text data

Output

15	4	25	29	22

Numerical Data

Outline

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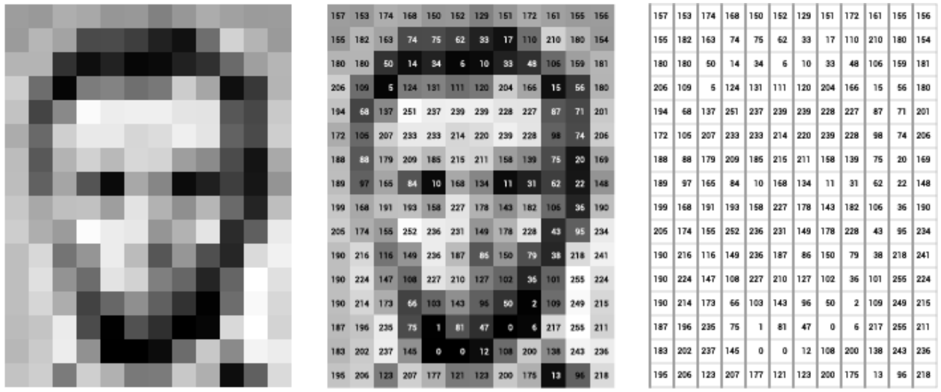
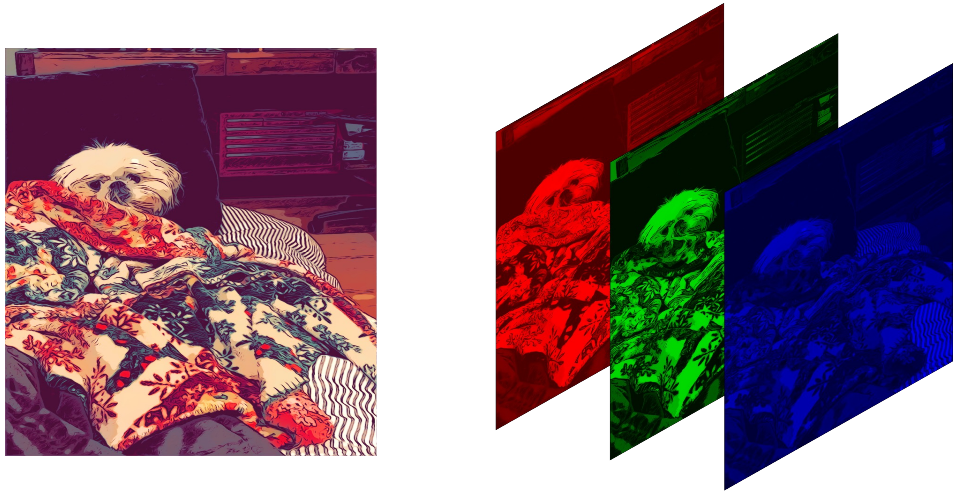
Image in Computer Vision

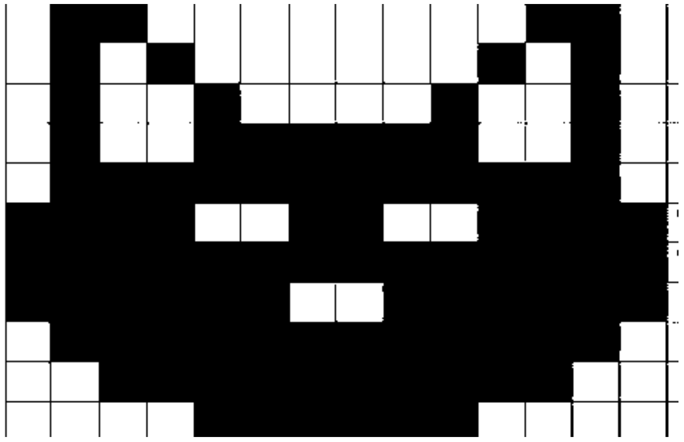


157	153	174	168	150	152	129	151	172	161	155	156
155	182	163	74	75	62	33	17	110	210	180	154
180	180	50	14	34	6	10	33	48	106	159	181
206	109	5	124	131	111	120	204	166	15	56	180
194	68	137	251	237	239	239	228	227	87	71	201
172	105	207	233	233	214	220	239	228	98	74	206
188	88	179	209	185	215	211	158	139	75	20	169
189	97	165	84	10	168	134	11	31	62	22	148
199	168	191	193	158	227	178	143	182	106	36	190
205	174	155	252	236	231	149	178	228	43	95	234
190	216	116	149	236	187	86	150	79	38	218	241
190	224	147	108	227	210	127	102	36	101	255	224
190	214	173	66	103	143	96	50	2	109	249	215
187	196	235	75	1	81	47	0	6	217	255	211
183	202	237	145	0	0	12	108	200	138	243	236
195	206	123	207	177	121	123	200	175	13	96	218

Grayscale image

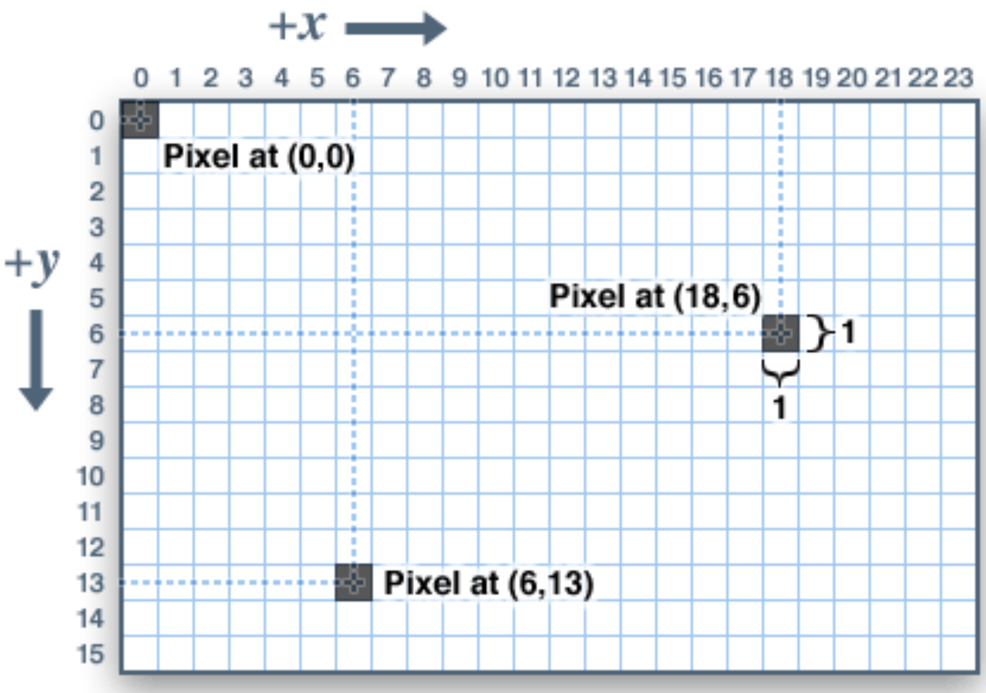


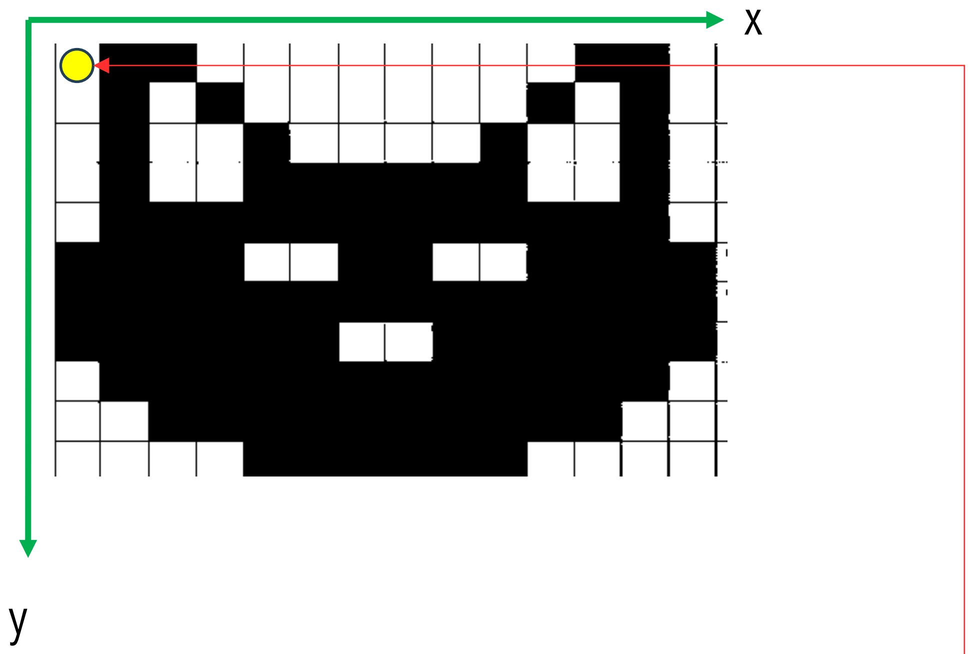
Color image



Binary Image

Image Coordinate System: Grayscale Image





Read image

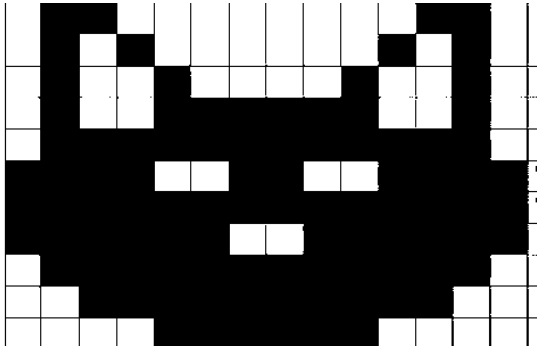
```
import cv2
from google.colab.patches import cv2_imshow

cat_image = cv2.imread("/content/binary_cat.png",0)
cv2_imshow(cat_image)
```

Access pixel at a specific location

```
x = 0
y = 0
pixel_value = cat_image[0][0]
print(pixel_value)
```

```
255
```

Image Histogram: Dictionary



Histogram Algorithm

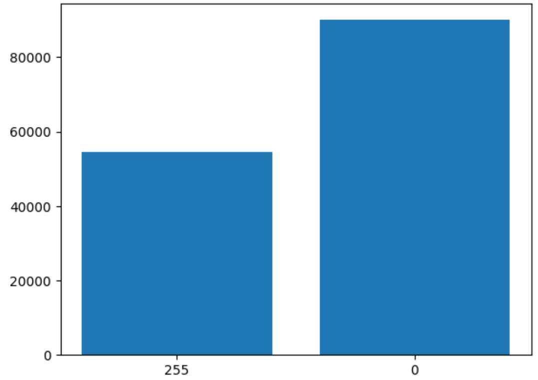
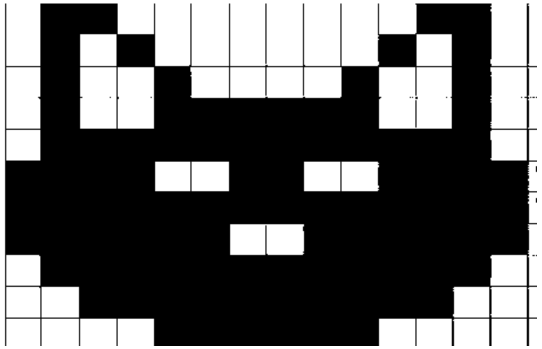
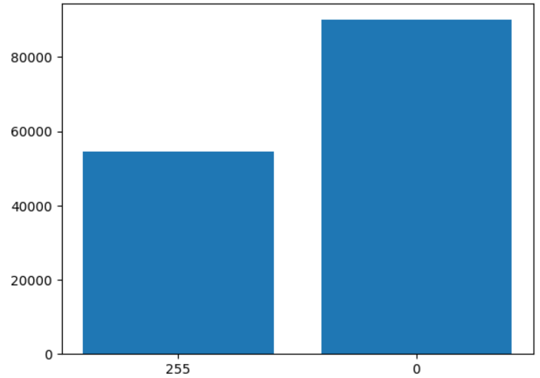


Image Histogram: Dictionary



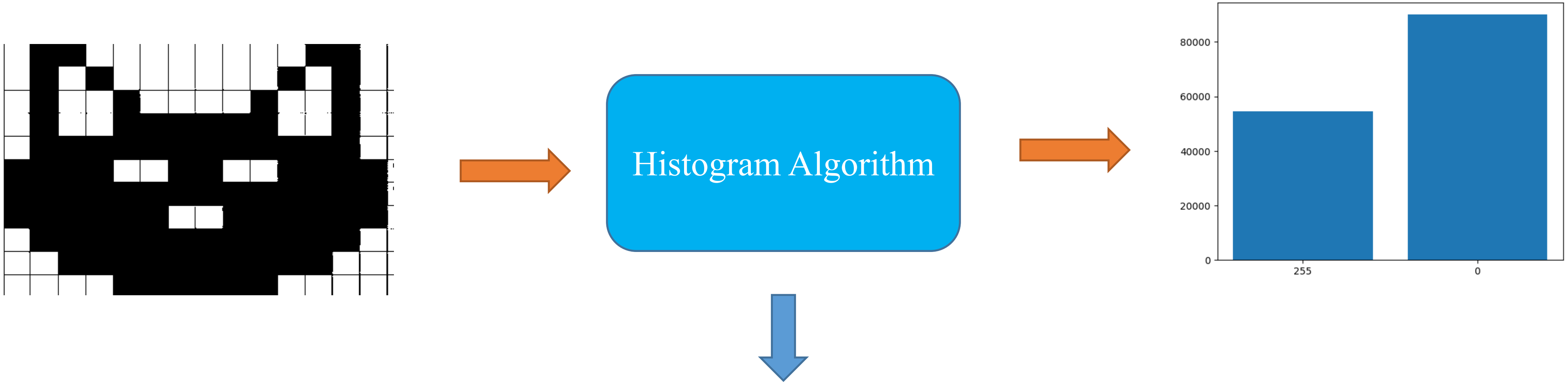
Histogram Algorithm



255	0	255	255	255	0	255
255	0	0	255	255	0	255
255	0	0	0	0	0	255
0	0	255	255	255	0	0
0	0	255	255	255	0	0
255	0	255	255	255	0	255
255	255	0	0	0	255	255

```
cat_image = cv2.imread("/content/binary_cat.png",0)
cv2_imshow(cat_image)
```

Image Histogram: Dictionary




```
counts = dict()
height, width = cat_image.shape
for row in range(height):
    for col in range(width):
        counts[cat_image[row,col]] = counts.get(cat_image[row,col],0) + 1
```

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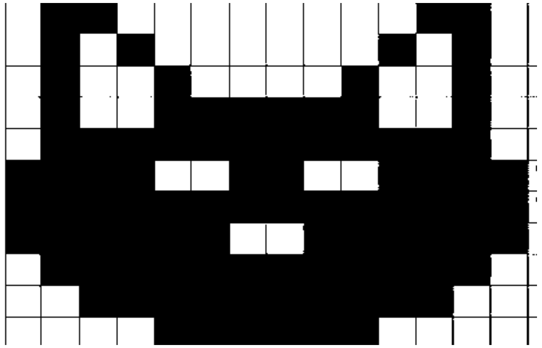
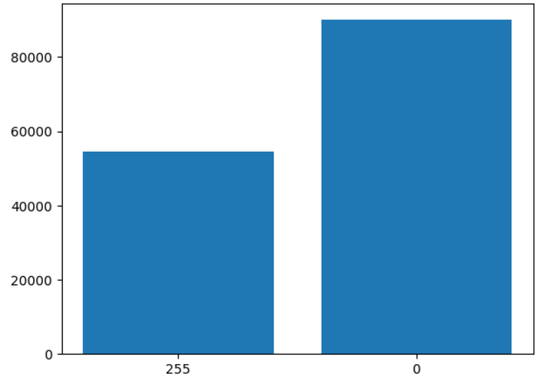
Image Histogram: Dictionary



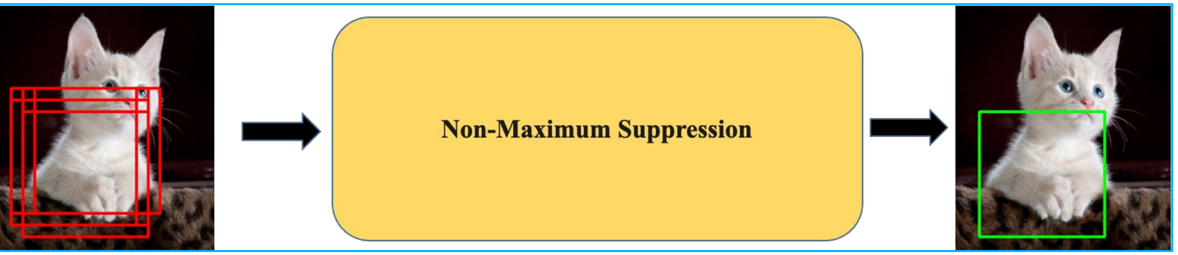
Histogram Algorithm



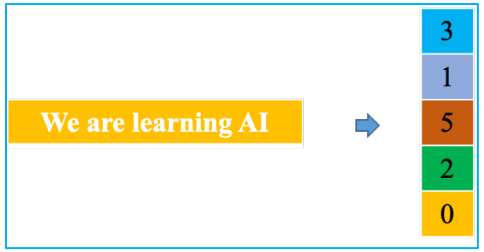

```
import matplotlib.pyplot as plt

names = list(counts.keys())
values = list(counts.values())

plt.bar(range(len(counts)), values, tick_label=names)
plt.show()
```

Summary

Data Structure	Ordered	Mutable	Constructor	Example
List	Yes	Yes	[] or list()	[5. 7, ‘yes’, 5.7]
Tuple	Yes	No	() or tuple()	(5.7, ‘yes’, 5.7)
Set	No	Yes	{} or set()	{5.7, ‘yes’ }
Dictionary	No	Yes	{} or dict{}	{‘key’: value}



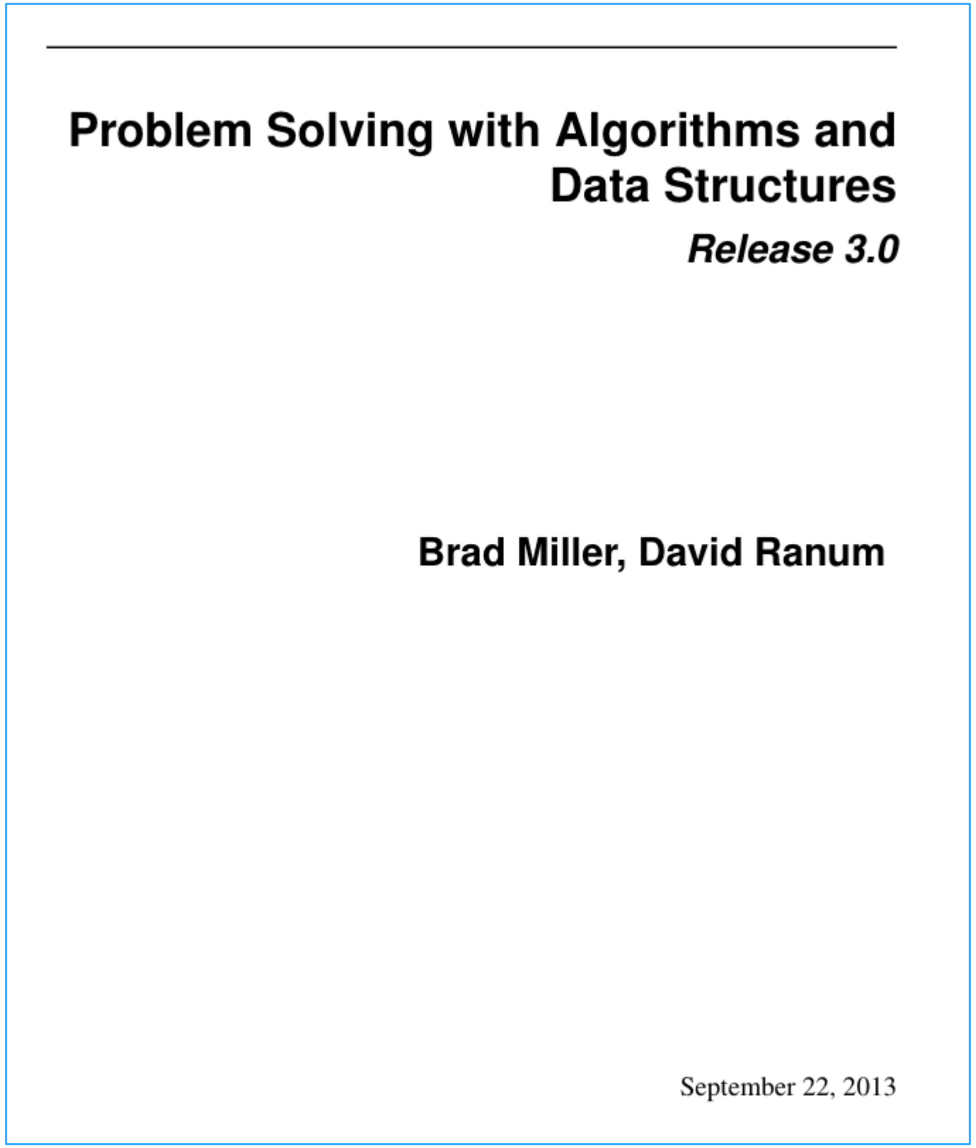
IOU, NMS using Tuple



Text embedding using Set

Histogram using Dictionary

References



Problem Solving with Algorithms and Data Structures
Release 3.0

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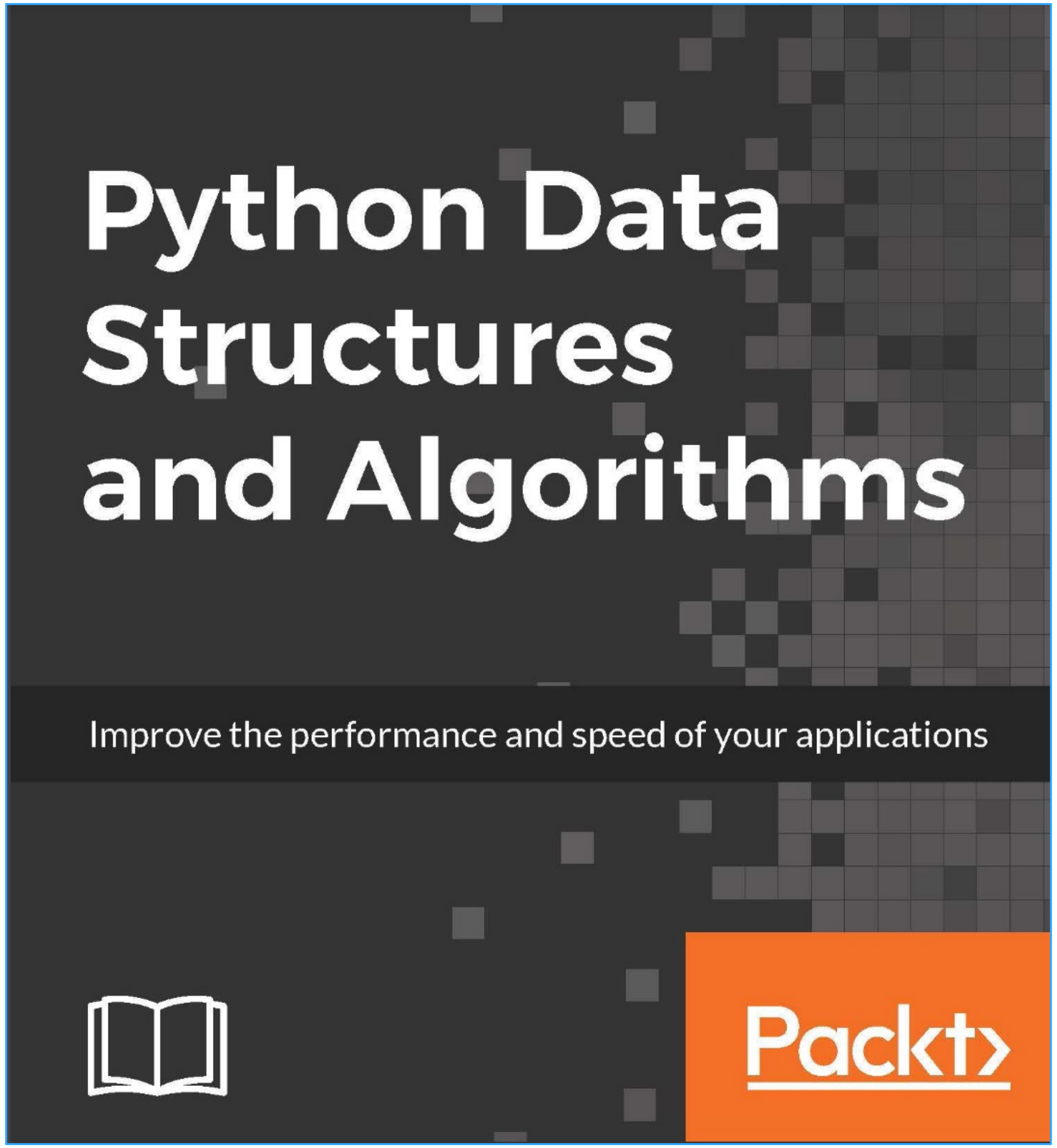
September 22, 2013

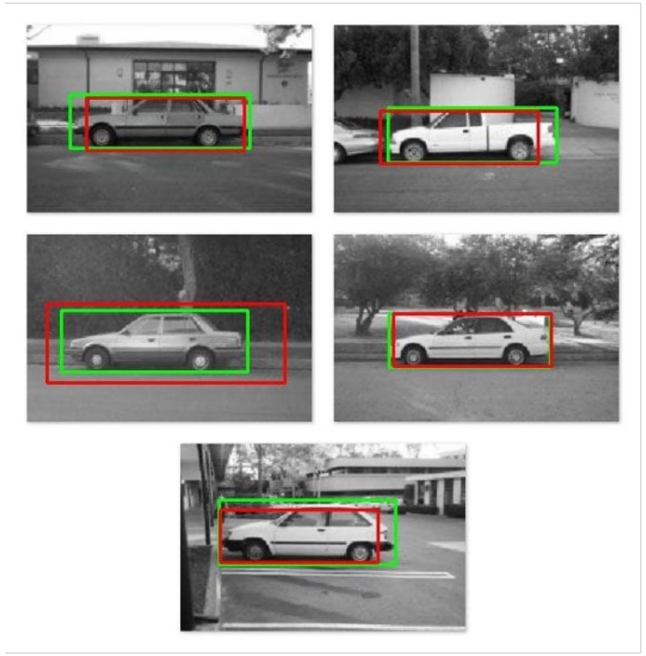


Outline

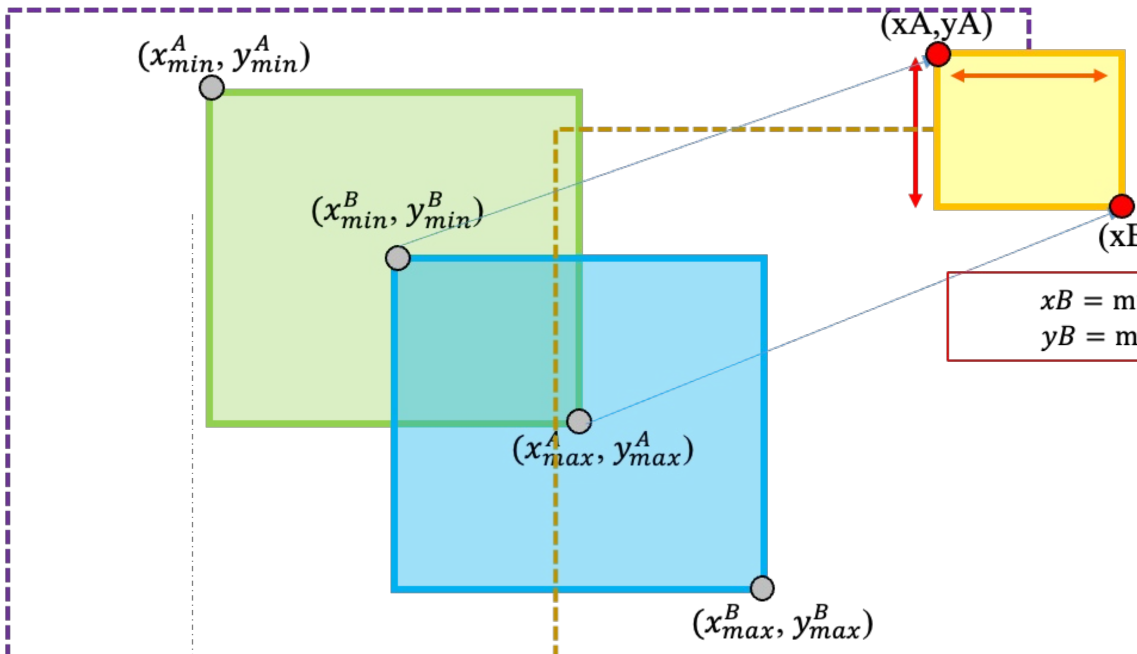
- What is a Tuple in Python
- What is a Set in Python
- What is a Dictionary in Python
- Quizz
- Text Classification using Set
- Image Histogram using Dictionary
- Further Reading


READ
MORE

IOU Calculation Using Tuple



$$xA = \max(x^A_{min}, x^b_{min}) = x^b_{min}$$
$$yA = \max(y^A_{min}, y^b_{min}) = y^b_{min}$$

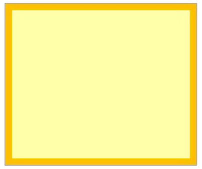


$$xB = \min(x^A_{max}, x^b_{max}) = x^A_{max}$$
$$yB = \min(y^A_{max}, y^b_{max}) = y^A_{max}$$

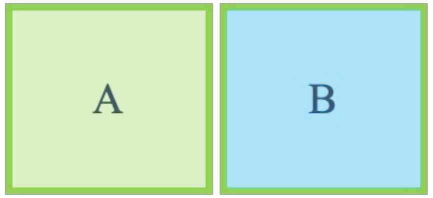
$$W = xB - xA$$

$$A \cap B = W * H$$

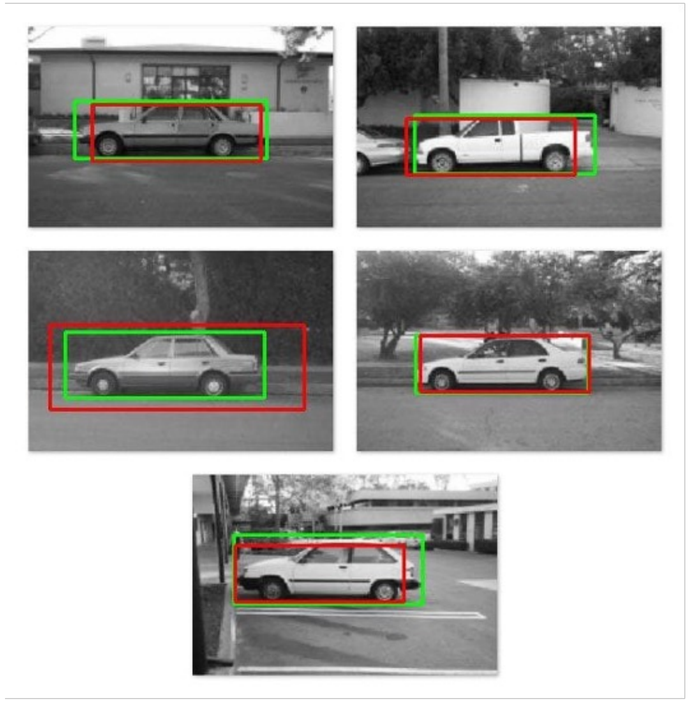
$$H = yB – yA$$



Area



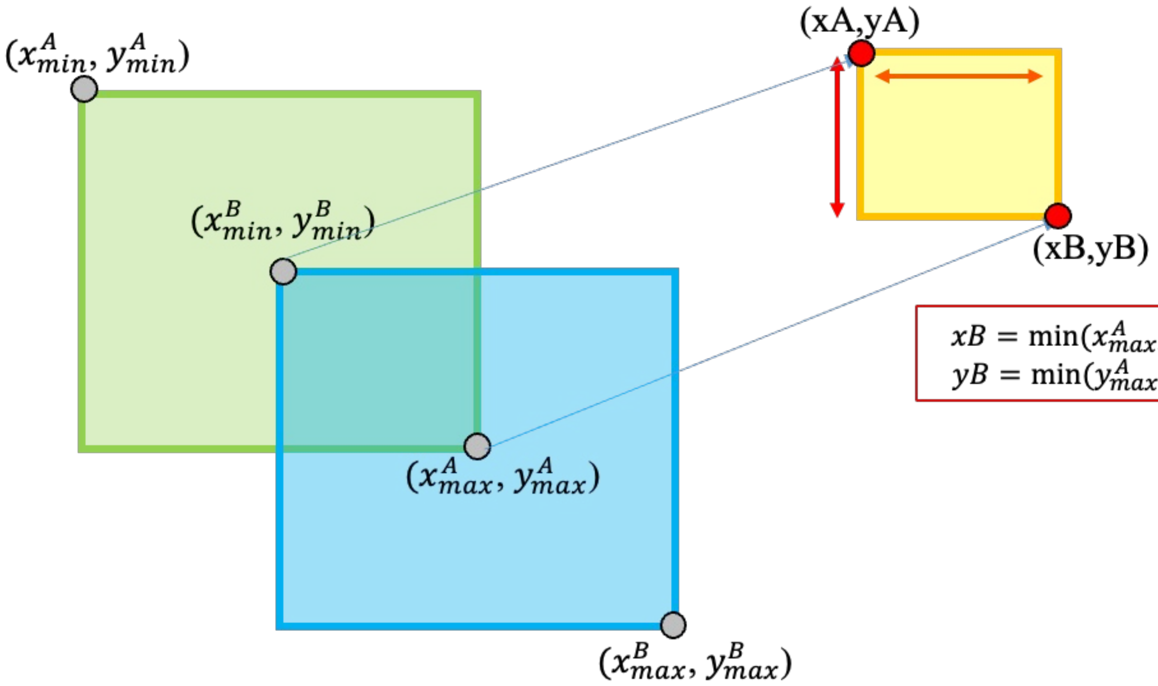
IOU Calculation Using Tuple



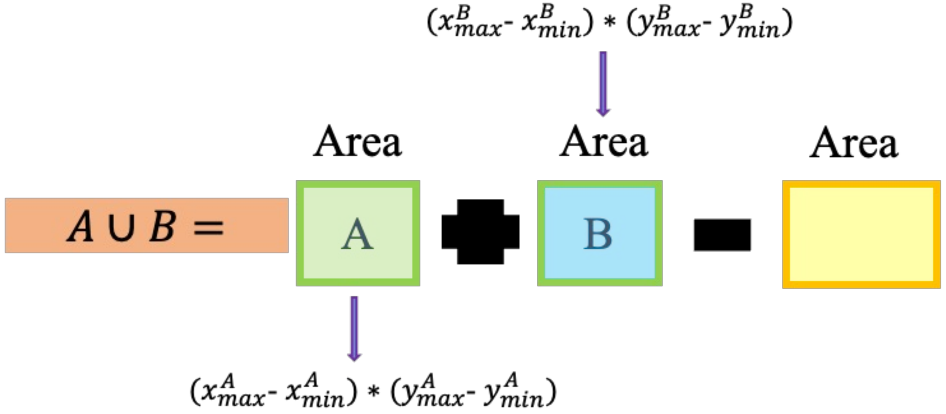
Giả sử ta có 2 boxes với thông tin như sau:

- **Box A** có tọa độ $(x^A_{min}, y^A_{min}, x^A_{max}, y^A_{max})$ => Tuple
- **Box B** có tọa độ $(x^B_{min}, y^B_{min}, x^B_{max}, y^B_{max})$ => Tuple

$$xA = \max(x^A_{min}, x^b_{min}) = x^b_{min}$$
$$yA = \max(y^A_{min}, y^b_{min}) = y^b_{min}$$



$$xB = \min(x^A_{max}, x^b_{max}) = x^A_{max}$$
$$yB = \min(y^A_{max}, y^b_{max}) = y^A_{max}$$



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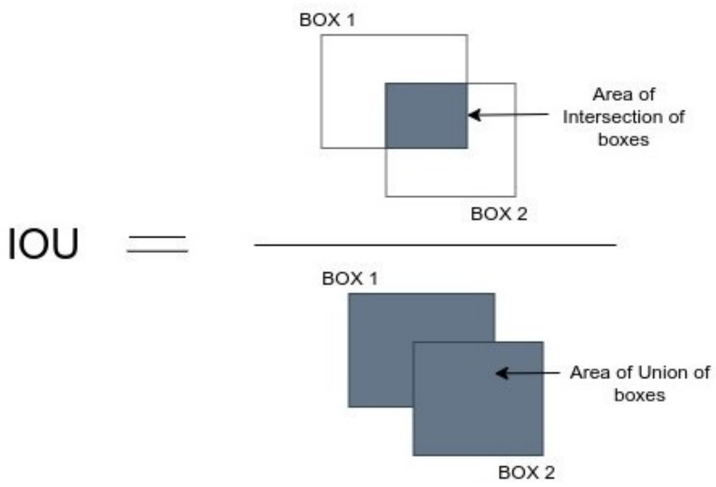
IOU Calculation Using Tuple



$$\frac{|A \cap B|}{|A \cup B|}$$

Tính diện tích của 2 box A, B

area1 = $(x^A_{max} - x^A_{min}) * (y^A_{max} - y^A_{min})$

area2 = $(x^B_{max} - x^B_{min}) * (y^B_{max} - y^B_{min})$

Tìm tọa độ của vùng giao nhau (Intersection)

$$xA = \max(x^A_{min}, x^b_{min}) = x^b_{min}$$
$$yA = \max(y^A_{min}, y^b_{min}) = y^b_{min}$$

$$xB = \min(x^A_{max}, x^b_{max}) = x^A_{max}$$
$$yB = \min(y^A_{max}, y^b_{max}) = y^A_{max}$$

Tính diện tích vùng giao nhau

$W = xB - xA$

H = yB – yA

intersection_area = w*h

Tính diện tích phần hợp nhau

union_area = area1 + area2 - intersection_area

Dựa trên phần giao và phần hợp để tính IoU

IoU = intersection_area / union_area

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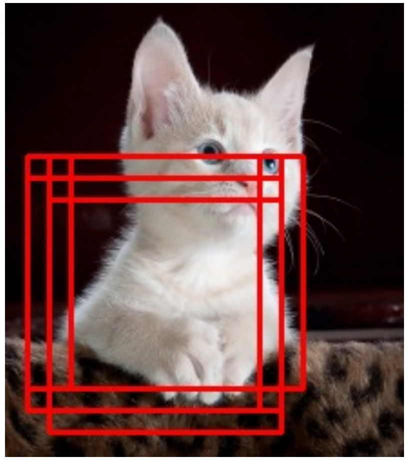
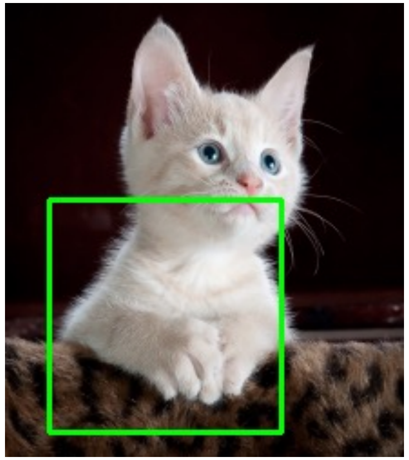
Non-Maxima Suppression: List & Tuple



Non Maximum Suppression



List | Tuple

```
# import the necessary packages
import numpy as np
import cv2
from google.colab.patches import cv2_imshow

boundingBoxes = np.array([
  (12, 84, 140, 212, 0.3),
  (24, 84, 152, 212, 0.4),
  (36, 84, 164, 212, 0.5),
  (12, 96, 140, 224, 0.6),
  (24, 96, 152, 224, 0.7),
  (24, 108, 152, 236, 0.8)])
```

Algorithm

Step 1 : Select the prediction **S** with highest confidence score and remove it from **P** and add it to the final prediction list **keep**. (**keep** is empty initially).

Step 2 : Now compare this prediction **S** with all the predictions present in **P**. Calculate the IoU of this prediction **S** with every other predictions in **P**. If the IoU is greater than the threshold **thresh_iou** for any prediction **T** present in **P**, remove prediction **T** from **P**.

Step 3 : If there are still predictions left in **P**, then go to **Step 1** again, else return the list **keep** containing the filtered predictions.